

Question	Answer	Mark	Mark scheme	Additional guidance
22	21	P1	for a relevant probability, eg $P(\text{green}) = \frac{x}{2x+3}$ or $P(\text{blue}) = \frac{x+3}{2x+3}$	the number of green and blue pens could be $x - 3$ and x or equivalent probabilities must be in an algebraic form in a single variable This is an exception using replacements. No further credit is available
		P1	for a relevant product, eg. $\frac{x}{2x+3} \times \frac{x-1}{2x+2}$ or $\frac{x+3}{2x+3} \times \frac{x+2}{2x+2}$ OR $\left(\frac{x}{x+3}\right)^2 + \left(\frac{x+3}{2x+3}\right)^2 = \frac{27}{75}$	
		P1	forms an appropriate equation, eg. $\frac{x}{2x+3} \times \frac{x-1}{2x+2} + \frac{x+3}{2x+3} \times \frac{x+2}{2x+2} = \frac{27}{55}$	
		P1	(dep P3) process to reduce equation to $ax^2 + bx + c = 0$ eg. $x^2 - 25x + 84 = 0$	
		P1	process to solve quadratic equation eg. $(x - 21)(x - 4) = 0$	
		A1	cao	